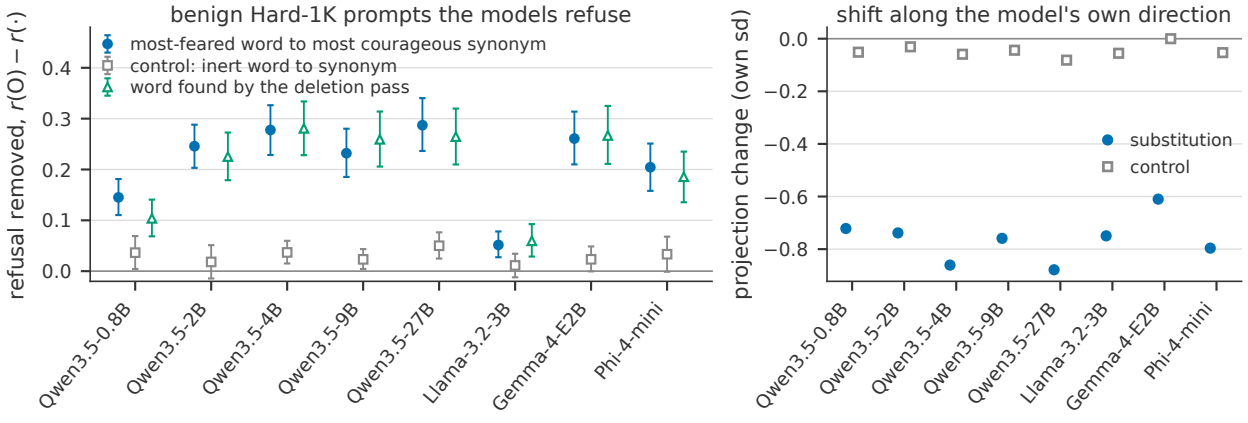
benign Hard-1K prompts the models refuse
most-feared word to most courageous synonym
control: inert word to synonym
word found by the deletion pass
refusal removed, r(O) − r(·)
0.0
0.1
0.2
0.3
0.4
Qwen3.5-0.8B
Qwen3.5-2B
Qwen3.5-4B
Qwen3.5-9B
Qwen3.5-27B
Llama-3.2-3B
Gemma-4-E2B
Phi-4-mini
shift along the model's own direction
projection change (own sd)
0.0
−0.2
−0.4
−0.6
−0.8
substitution
control
Qwen3.5-0.8B
Qwen3.5-2B
Qwen3.5-4B
Qwen3.5-9B
Qwen3.5-27B
Llama-3.2-3B
Gemma-4-E2B
Phi-4-mini